



Modelling Financial Markets

Chapter 12: Options and the Volatility Surface



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Acronyms used in this material

- ATM: At-The-Money
- COVID-19: COronaVirus Disease 2019
- CRR: Cox–Ross–Rubinstein (binomial tree)
- DVOL: Deribit Volatility Index
- ETF: Exchange-Traded Fund
- GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- GJR-GARCH: Glosten–Jagannathan–Runkle GARCH
- ITM: In-The-Money
- OTM: Out-of-The-Money
- SVI: Stochastic Volatility Inspired (parameterisation)
- US: United States
- UTC: Coordinated Universal Time (Temps Universel Coordonné)
- VIX: CBOE Volatility Index
- VIX3M: Cboe 3-Month Volatility Index
- VIX9D: Cboe 9-Day Volatility Index
- VRP: Variance Risk Premium
- VVIX: Cboe VIX of VIX Index
- XFG: Quantlet prefix of Applied Quantitative Finance (Härdle, Chen & Overbeck)

Outline

Motivation

Pay-offs and Put–Call Parity

The Binomial Tree

Black–Scholes and the Greeks

Delta Hedging in Practice

Implied Volatility and the Smile

Bitcoin Options and the Volatility Surface

The VIX and the Variance Risk Premium

Options with Zero Days to Expiry

Summary

References

Why Options?

What an option does

- An **option** gives the right, not the obligation, to buy or sell an asset at a fixed price
- It pays only in some states of the world: a bet on the **distribution**, not only on the direction
- Option prices reveal what the market expects about volatility and crashes

What we do in this chapter

- Price options: parity, the binomial tree, Black–Scholes and the sensitivities
- Hedge them and measure what the hedge leaves behind
- Read implied volatility: smile, surface, VIX, variance risk premium
- New markets: options with zero days to expiry and Bitcoin options

Chicago, 1973: Listed Options Begin



Trading floor of the Chicago Board of Trade, 2006

Photo: [Richie Diesterheft \(2006\)](#); CC BY 2.0; [Wikimedia](#)

[Commons](#)

- The Chicago Board Options Exchange (Cboe) opened in 1973 as the first exchange for standardised, listed options [MacKenzie & Millo \(2003\)](#)
 - ▶ Before: options were negotiated one by one, over the counter
- The same year: the Black–Scholes and Merton pricing papers [Black & Scholes \(1973\)](#), [Merton \(1973\)](#)
- Traders soon used the formula on the floor: theory changed the market it described [MacKenzie & Millo \(2003\)](#)
- Textbooks: [Franke, Härdle & Hafner \(2019\)](#), [Härdle, Chen & Overbeck \(2017\)](#), [Gatheral \(2006\)](#)

The Formula and Its Authors



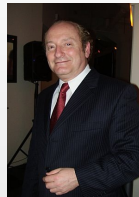
Fischer Black (1938–1995)

Photo: Dalmatine at English Wikipedia;
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Myron Scholes (2008)

Photo: Nobellaureatesphotographer (2008);
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Robert C. Merton (2006)

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- Key idea: a portfolio of the stock and cash can **replicate** the option, so its price follows from no-arbitrage [Black & Scholes \(1973\)](#)
- Scholes and Merton received the [1997 Nobel Memorial Prize in Economic Sciences](#); Black had died in 1995

Data Used in This Chapter

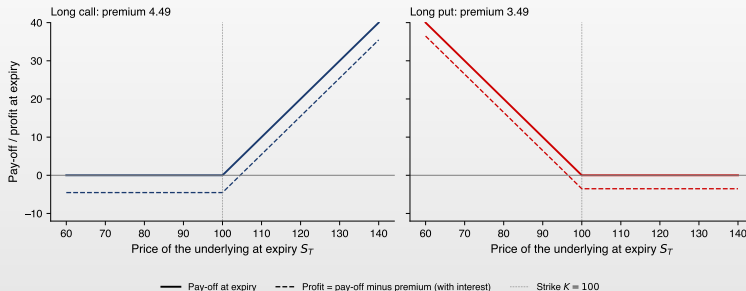
Series	What it measures	Period	Owner
S&P 500	US equity index, daily close	1990–2026	S&P Dow Jones Indices
SPY	ETF on the S&P 500: open and close	1993–2026	State Street
VIX	30-day implied volatility of the S&P 500	1990–2026	Cboe
VIX9D, VIX3M	the same over 9 days and 3 months	2011 / 2007–2026	Cboe
VVIX	implied volatility of the VIX	2006–2026	Cboe
Bitcoin options	all listed contracts, one dated snapshot	22 Sep 2026	Deribit
DVOL	30-day implied volatility of Bitcoin	2021–2026	Deribit

- Daily data up to 18 September 2026; the option snapshot was taken on 22 September 2026, 07:51 UTC
- Implied volatility is quoted in annualised %; “vol points” = percentage points of volatility

The Vocabulary of Options

- **Call**: the right to **buy** the underlying at the strike K ; **put**: the right to **sell** at K
 - ▶ **Underlying**: the asset the option is written on (price S_t); **premium**: the price paid today for the right
- **Expiry** T : the last day of the right; $\tau = T - t =$ time to expiry, in years
- **European** options can be exercised only at T ; **American** options at any time up to T
- **Moneyness** compares K with S : at-the-money (ATM) $K \approx S$; in-the-money (ITM) when exercise would pay now; out-of-the-money (OTM) otherwise
 - ▶ Log-moneyness $k = \ln(K/F)$, with F the forward price of the underlying for expiry T
- Pay-offs at expiry: call $\max(S_T - K, 0)$, put $\max(K - S_T, 0)$

Pay-off and Profit at Expiry

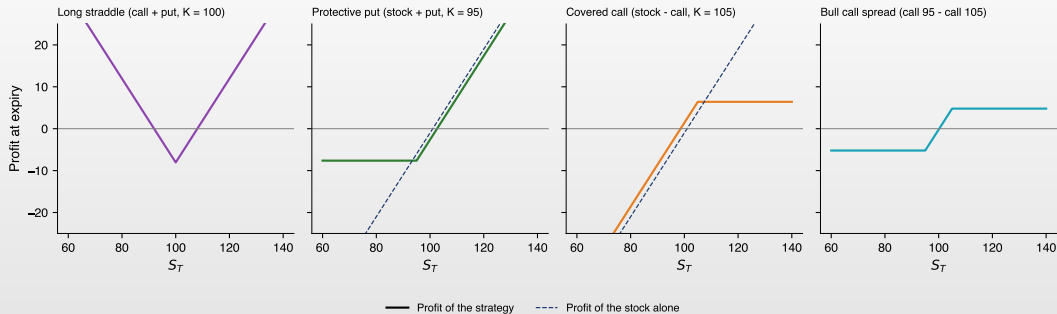


□ $K = 100$, three months, volatility 20%, interest rate 4%; premia from Black-Scholes (Section 4)

Reading the Pay-off Diagrams

- Long call: premium 4.49; the loss is capped at the premium, the gain is unlimited
 - ▶ Break-even at expiry: $S_T = K + \text{premium with interest}$
- Long put: premium 3.49; the gain is capped at K (the price cannot fall below 0)
- Both pay-offs are **convex** (kinked at K): the holder gains from large moves in one direction
- The seller has the mirror image: a small certain gain (the premium) against a large possible loss

Four Classic Strategies



□ Profit at expiry, same parameters; stock bought at 100 and financed at the risk-free rate

What Each Strategy Bets On

- ▣ **Straddle** (call + put at the same K): gains if the price moves a lot either way: a bet on **volatility**
- ▣ **Protective put** (stock + put): insurance; the loss is floored, the upside kept, at the cost of the premium
- ▣ **Covered call** (stock – call): sells the upside above K for the premium: income in calm markets
- ▣ **Bull call spread** (call 95 – call 105): a cheap directional bet with capped gain and loss
- ▣ Options let an investor choose the **shape** of the pay-off, not only its direction

Put–Call Parity

- Two portfolios with the same pay-off at T (European options, same K and T , no dividends):
 - ▶ A: one call + cash Ke^{-rT} (worth K at T) $\Rightarrow \max(S_T, K)$
 - ▶ B: one put + one share $\Rightarrow \max(K, S_T)$
- Same pay-off in every state, so the same price today, otherwise there is an arbitrage:
 - ▶ $C_t - P_t = S_t - Ke^{-rT}$ (with dividend yield q : S_te^{-qT} instead of S_t)
- No model for S is needed: parity uses only no-arbitrage [Merton \(1973\)](#)
- Consequence: a call and a put with the same K and T have the **same** implied volatility (Section 6)

Worked Example: Parity and an Arbitrage

- $S = K = 100$, six months, $r = 4\%$, volatility 20%: $C = 6.63$, $P = 4.65$
 - ▶ $C - P = 1.98$ and $S - Ke^{-r\tau} = 100 - 98.02 = 1.98$: parity holds
- Suppose the market quotes the put at 4.00 instead of 4.65: $C - P$ is too large
 - ▶ Sell the call, buy the put, buy the share, borrow $Ke^{-r\tau}$: cash today $= C - P - S + Ke^{-r\tau} > 0$
 - ▶ At expiry the position is worth exactly 0 in every state: a riskless profit
- No-arbitrage bounds: $\max(S - Ke^{-r\tau}, 0) \leq C \leq S$ and $\max(Ke^{-r\tau} - S, 0) \leq P \leq Ke^{-r\tau}$
 - ▶ An American call on a stock without dividends is never exercised early: it is worth at least $S - Ke^{-r\tau} > S - K$

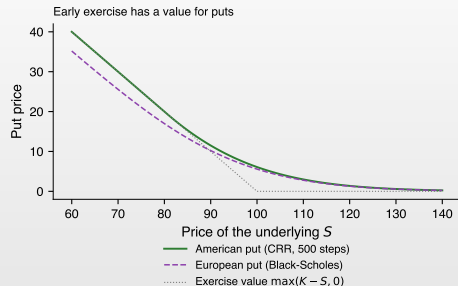
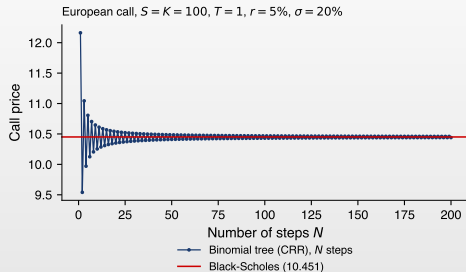
One Period: Replicating a Call

- Today S ; after one period uS or dS , with $d < R < u$, $R = e^{r\Delta t}$ the gross risk-free return
 - ▶ The option pays C_u or C_d
- Buy Δ shares and invest B in cash so that the portfolio pays C_u and C_d :
 - ▶ $\Delta uS + BR = C_u, \quad \Delta dS + BR = C_d \Rightarrow \Delta = \frac{C_u - C_d}{(u - d)S}, \quad B = \frac{uC_d - dC_u}{(u - d)R}$
- No arbitrage: the option costs what the replicating portfolio costs, $C = \Delta S + B$
- Model: the **Cox–Ross–Rubinstein** tree (CRR) [Cox, Ross & Rubinstein \(1979\)](#); worked numbers in Seminar 12, A3

Risk-Neutral Probability

- Rewrite the price as a discounted expectation: $C = \frac{q C_u + (1 - q) C_d}{R}$, $q = \frac{R - d}{u - d}$
 - ▶ q is the probability under which the stock earns exactly the risk-free rate: $q uS + (1 - q) dS = RS$
- The **real** probability of an up move does not enter the price: investors' risk aversion is already in S
- q is a pricing device, not a forecast: "risk-neutral" prices embed risk premia (Section 7)
- Many periods (CRR): $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$; value the tree backwards from T
 - ▶ American option: at every node take $\max(\text{continuation value, exercise value})$

From the Tree to Black-Scholes



□ Left: CRR price of a European call for $N = 1, \dots, 200$ steps; right: American vs European put

Reading the Binomial Results

- The tree converges to Black–Scholes (10.451) as N grows: $N = 50$: 10.411; $N = 200$: 10.441; $N = 1000$: 10.449
 - ▶ Convergence oscillates: $N = 10$ gives 10.253, $N = 11$ gives 10.611 (the strike falls on or between nodes)
- American put at $S = K = 100$: 6.090 vs European 5.574
 - ▶ Early-exercise premium 0.516: deep in the money it pays to receive K now and earn interest
- The tree is the workhorse for American options; Black–Scholes is its limit for European ones
- Recap: replication $\Rightarrow$ price; risk-neutral probability $\Rightarrow$ price as a discounted expectation

The Black–Scholes Model

- Price follows a **geometric Brownian motion**: $dS_t = \mu S_t dt + \sigma S_t dW_t$
 - ▶ W_t : Brownian motion, independent Normal increments $W_{t+h} - W_t \sim N(0, h)$
 - ▶ So log returns over τ follow the Normal distribution with variance $\sigma^2 \tau$: σ is the annualised volatility
- Assumptions: constant σ and r , continuous trading, no costs, short selling allowed, no arbitrage
- Continuous-time limit of the tree: the hedge Δ is adjusted at every instant [Black & Scholes \(1973\)](#), [Merton \(1973\)](#)
 - ▶ The drift μ drops out, exactly as the real probability did in the tree
- Chapter 11 derives these processes; here we use their consequences

The Black–Scholes Formula

- European call and put, no dividends (with yield q : replace S by $Se^{-q\tau}$):
 - ▶ $C = S\Phi(d_1) - Ke^{-r\tau}\Phi(d_2), \quad P = Ke^{-r\tau}\Phi(-d_2) - S\Phi(-d_1)$
 - ▶ $d_1 = \frac{\ln(S/K) + (r + \sigma^2/2)\tau}{\sigma\sqrt{\tau}}, \quad d_2 = d_1 - \sigma\sqrt{\tau},$ with Φ the standard Normal distribution function
- Reading the two terms:
 - ▶ $\Phi(d_2)$ = risk-neutral probability that the call ends in the money
 - ▶ $\Phi(d_1)$ = number of shares in the replicating portfolio (the delta)
- Five inputs: S, K, τ, r observed; σ is the only one not observed: it must be estimated or implied

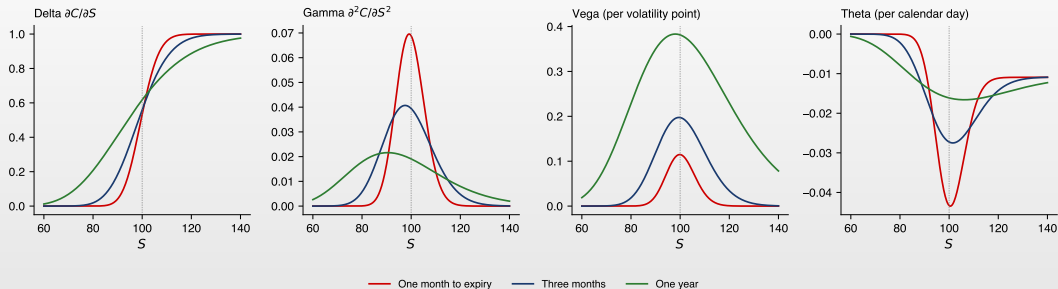
Worked Example: Pricing a Call

- ▣ $S = 100$, $K = 105$, three months ($\tau = 0.25$), $r = 4\%$, $\sigma = 16\%$
- ▣ $d_1 = \frac{\ln(100/105) + (0.04 + 0.0128) \cdot 0.25}{0.16 \cdot 0.5} = -0.4449$, $d_2 = -0.4449 - 0.08 = -0.5249$
- ▣ $\Phi(d_1) = 0.3282$, $\Phi(d_2) = 0.2998$, $Ke^{-r\tau} = 103.96$
- ▣ $C = 100 \cdot 0.3282 - 103.96 \cdot 0.2998 = 1.65$; parity gives $P = C - S + Ke^{-r\tau} = 5.61$
- ▣ The call is out of the money: a 0.2998 risk-neutral chance of exercise, priced at 1.65% of the share

The Greeks: Sensitivities of the Price

- ▣ **Delta** $\Delta = \partial C / \partial S = \Phi(d_1)$ (put: $\Phi(d_1) - 1$): shares needed to hedge one option
 - **Gamma** $\Gamma = \partial^2 C / \partial S^2 = \varphi(d_1) / (S\sigma\sqrt{\tau})$: how fast delta changes; φ = standard Normal density
- ▣ **Vega** $\mathcal{V} = \partial C / \partial \sigma = S\varphi(d_1)\sqrt{\tau}$: gain per unit of volatility (we quote it per 1 vol point)
- ▣ **Theta** $\Theta = \partial C / \partial t = -\frac{S\varphi(d_1)\sigma}{2\sqrt{\tau}} - rKe^{-r\tau}\Phi(d_2)$: time decay (we quote it per calendar day)
- ▣ **Rho** $\rho = \partial C / \partial r = K\tau e^{-r\tau}\Phi(d_2)$: sensitivity to the interest rate
- ▣ Call and put with the same K , T have the same gamma and vega (parity)

The Greeks Across Prices and Maturities



□ Call with $K = 100$, volatility 20%, $r = 4\%$, for one month, three months and one year to expiry

Reading the Greeks

- Close to expiry delta becomes a step: 0 below K , 1 above K
 - ▶ Gamma and theta concentrate at the money: the hedge must change fast and the option loses value fast
- Vega is largest for long maturities: long options are bets on volatility
- Example of the previous slide: $\Delta = 0.3282$, $\Gamma = 0.0452$, vega 0.1807 per vol point, theta -0.0193 per day
 - ▶ One more vol point adds 0.1807 to a price of 1.65: a 11% change
- Theta and gamma are linked: $\Theta + \frac{1}{2}\sigma^2 S^2 \Gamma + rS\Delta - rC = 0$ (the Black–Scholes equation)
 - ▶ Long gamma is paid for with time decay; short gamma earns it

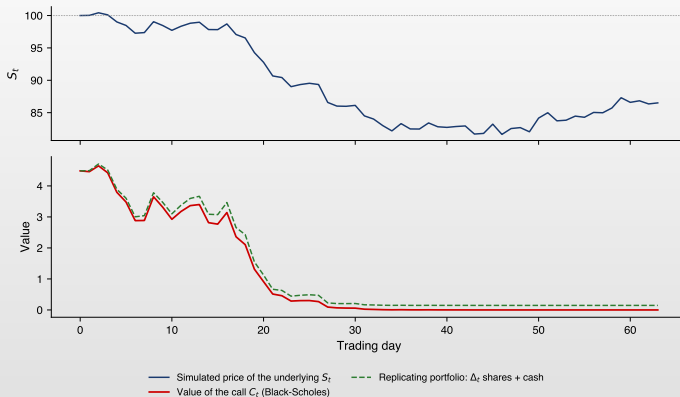
Recap: Black–Scholes

- Replication with a continuously adjusted delta gives a price that does not depend on μ
- The price depends on one unobserved number, σ
- The Greeks tell a trader what the position is exposed to: price (delta, gamma), volatility (vega), time (theta)
- Next: what happens when the hedge is adjusted only at discrete times, and when σ is wrong

Discrete Delta Hedging: the Algorithm

- Sell one call at the Black–Scholes price C_0 with volatility σ_{imp} ; buy Δ_0 shares; the rest is cash
- At each of N rebalancing dates: recompute Δ at the current price and time, buy or sell shares, update the cash with interest
- At expiry: **hedging error** = value of shares and cash minus the option pay-off
- Simulation: geometric Brownian motion, $\mu = 8\%$, $\sigma = 20\%$, three months, 20 000 paths
 - ▶ If the model is right and $\sigma_{imp} = \sigma$, the error vanishes as $N \rightarrow \infty$

One Path: the Option and Its Replica

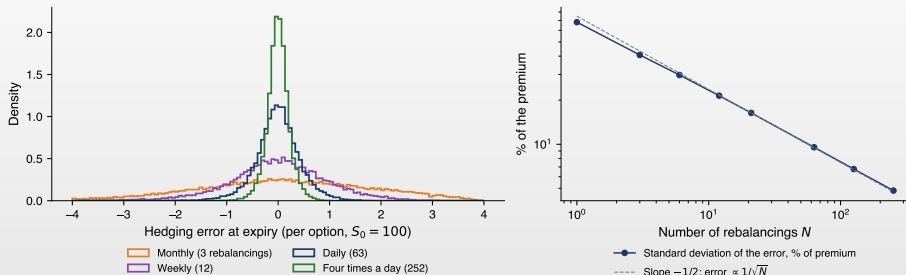


□ Daily rebalancing, 63 trading days; top: the simulated price; bottom: call value vs replicating portfolio

Reading the Replication Path

- ▣ The replicating portfolio follows the option value day by day, starting from the premium 4.49
- ▣ The price ends at 86.5, the call expires worthless; the portfolio ends at 0.15 instead of 0
- ▣ The small gap comes from rebalancing once a day instead of continuously
- ▣ The seller who hedges does not bet on direction: the result depends on how the price **moved**, not where it went

Hedging Error and Rebalancing Frequency



□ Left: distribution of the hedging error for four frequencies; right: its standard deviation as % of the premium

 MFM_ch12_hedge_error

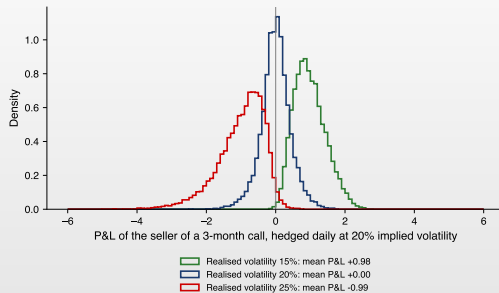
Reading the Hedging Error

- The mean error is close to 0 at every frequency: the premium is fair on average
 - ▶ Its standard deviation falls with N : monthly 40.8%, weekly 21.5%, daily 9.5%, four times a day 4.8% of the premium
- Log-log slope -0.48 : the error shrinks like $1/\sqrt{N}$; four times more trades halve it
- Worst 1% of paths: -2.68 with weekly and -1.17 with daily rebalancing, per option on a price of 100
- With transaction costs more trades cost more: the optimal frequency trades off error against cost

When Realised Volatility Differs from Implied

- Hedging at σ_{imp} when the price moves with volatility σ_{real} leaves a P&L (profit and loss) for the seller of
 - ▶
$$P\&L \approx \int_0^T \frac{1}{2} \Gamma_t S_t^2 \left(\sigma_{imp}^2 - \sigma_{real}^2 \right) dt$$
- The seller earns time decay and pays for every large move through gamma: a bet that realised volatility stays **below** implied
- The P&L is weighted by gamma: large moves near the strike and near expiry matter most
- First-order size: vega $\times$ (implied $-$ realised); here vega = 0.197 per vol point, so 5 vol points $\approx$ 0.99

Selling Volatility at 20%: Three Outcomes

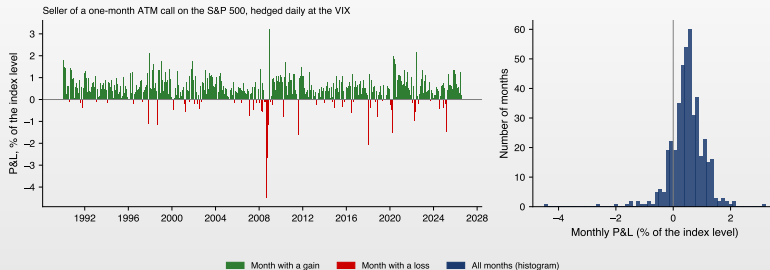


- Three-month call sold at 20% implied volatility, hedged daily; realised volatility 15%, 20% or 25%

Reading the Volatility Bet

- ▣ Realised 15%: mean P&L $+0.98$, positive on 99.8% of paths
- ▣ Realised 20%: mean $+0.00$: the premium was fair, only the discrete-hedging noise remains (s.d. 0.43)
- ▣ Realised 25%: mean -0.99 , positive on only 2.2% of paths
- ▣ Each result is close to $\text{vega} \times 5 \text{ vol points } (\pm 0.99)$: a delta-hedged option is a clean bet on volatility
- ▣ Next: which of the three worlds did the S&P 500 live in?

Thirty-Six Years of Selling S&P 500 Volatility



- Every 21 trading days: sell a one-month at-the-money call priced with the VIX, hedge it daily at the same volatility; P&L in % of the index; interest and dividends ignored

 MFM_ch12_delta_hedged_sp500

Reading the Delta-Hedged History

- ▣ 439 months from 1990-01: mean P&L 0.48% of the index per month, Newey–West $t = 14.0$ Newey & West (1987)
 - ▶ Positive in 84% of months; mean premium collected 2.23%
- ▣ The VIX exceeded the realised volatility of the month in 85% of months (means 19.4 vs 15.4)
- ▣ Worst month -4.51% (2008-09); skewness -1.53 : many small gains, a few large losses
- ▣ Option sellers earn a **volatility risk premium**: insurance against turbulence is expensive Bakshi & Kapadia (2003)

Recap: Hedging

- ▣ Discrete hedging leaves an error that shrinks like $1/\sqrt{N}$
- ▣ A delta-hedged option earns implied minus realised variance, weighted by gamma
- ▣ On the S&P 500, implied volatility has been higher than realised most of the time: sellers were paid, with crash risk
- ▣ Hedging needs a volatility: the next question is which one the market uses

Implied Volatility

- **Implied volatility** σ_{imp} : the σ that makes the Black–Scholes price equal the market price
 - ▶ $BS(S, K, \tau, r, \sigma_{imp}) = C^{market}$; unique because vega > 0 : the price increases in σ
- Newton's method: $\sigma_{k+1} = \sigma_k - \frac{BS(\sigma_k) - C^{market}}{\mathcal{V}(\sigma_k)}$
 - ▶ Example: $S = K = 100$, three months, $r = 2\%$, market price 4.50; start at $\sigma_0 = 20.00\%$
 - ▶ $BS(\sigma_0) = 4.2322$, $\mathcal{V} = 19.848 \Rightarrow \sigma_1 = 21.3495\%$; the next step changes only the 6th decimal: $\sigma_{imp} = 21.3495\%$
- Traders quote options in implied volatility, not in money: it is comparable across strikes, maturities and assets
- If Black–Scholes were right, all options on one asset would have the **same** implied volatility

After October 1987: the Smile Appears



Amsterdam stock exchange after the crash, 21 October
1987

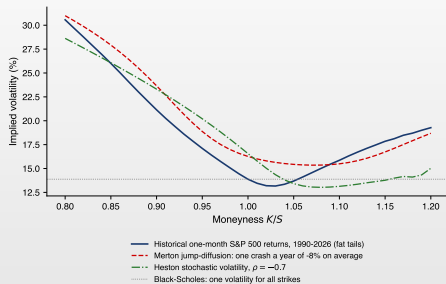
Photo: [Bart Molendijk / Anefo \(1987\)](#); CC0; [Wikimedia Commons](#)

- Before 1987 index options had a nearly flat implied volatility across strikes [Rubinstein \(1994\)](#)
 - ▶ Since the crash, low-strike puts are much more expensive: the skew
- Option prices in 1987 already showed fear of a crash before it happened [Bates \(1991\)](#)
- **Smile**: implied volatility as a function of K ; **term structure**: as a function of T ; **surface**: both
- Crash risk on the Bucharest Stock Exchange: [Pele & Mazurencu-Marinescu \(2012\)](#)

Why Do Smiles Exist?

- **Fat tails:** large moves are more frequent than the Normal distribution allows (Chapter 1)
 - ▶ Deep OTM options pay only after large moves, so they are worth more than Black–Scholes says
- **Leverage and volatility feedback:** when prices fall, volatility rises [Christie \(1982\)](#)
 - ▶ Negative skewness of returns: the left tail is longer, so low-strike puts are dearer
- **Crash fear and demand for insurance:** investors pay a premium for puts [Bates \(1991\)](#)
 - ▶ Intermediaries who sell them need compensation for crash risk they cannot hedge
- Models: jumps [Merton \(1976\)](#), stochastic volatility [Heston \(1993\)](#), [Hull & White \(1987\)](#), local volatility [Gatheral \(2006\)](#)

Three Sources of a Smile

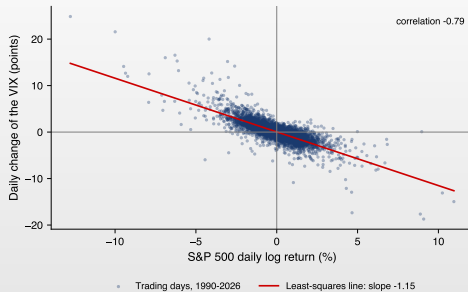


- One-month implied volatility from: the historical distribution of S&P 500 monthly returns (made risk-neutral), a jump model and a stochastic-volatility model

Reading the Model Smiles

- ▣ Historical S&P 500: 9 220 overlapping one-month returns, skewness -1.27 , excess kurtosis 6.8
 - ▶ Implied volatility 30.6% at $K/S = 0.8$, 13.9% at the money, 19.3% at 1.2: fat tails alone create a skewed smile
- ▣ Merton jumps (one crash of about -8% a year): 31.0% vs 16.3%: jumps steepen short-dated smiles
- ▣ Heston with $\rho = -0.7$: 28.6% vs 16.6% vs 13.1%: negative correlation between price and variance tilts the smile into a skew
- ▣ Market smiles are steeper still: the risk premium for crashes adds to the physical skew

The Leverage Effect in the Data



- Daily S&P 500 log return vs the change of the VIX on the same day, 1990–2026

Reading the Leverage Effect

- ▣ Correlation -0.79 over 9 239 days: implied volatility rises when the market falls
- ▣ Slope -1.15 VIX points per 1% return; steeper on down days (-1.39) than on up days (-1.01)
 - ▶ Mean VIX change: $+0.88$ on down days, -0.76 on up days
- ▣ This is the $\rho < 0$ of the Heston model and the asymmetry of GJR-GARCH (Chapter 5)
- ▣ A long position in low-strike puts gains from both the fall and the rise in volatility: this double hedge is why they are expensive

Models of the Smile

Model	Idea	Fits best
Jumps Merton (1976)	Poisson jumps added to the Brownian motion	steep short-dated smiles
Stochastic volatility Heston (1993)	variance follows its own mean-reverting process, correlated with S	skew and term structure of longer maturities
Local volatility Gatheral (2006)	$\sigma(S, t)$ read off the whole surface (Dupire's formula)	matches every quoted price today, exotic options
Parametric smile, SVI Gatheral & Jacquier (2014)	five parameters per expiry, no dynamics	interpolation and quoting (next section)

- Chapter 11 introduces the jump and stochastic-volatility processes; [Franke, Härdle & Hafner \(2019\)](#) and [Čížek, Härdle & Weron \(2011\)](#) give the implied-volatility surface in detail

Recap: Implied Volatility

- ▣ Implied volatility is the market price of an option translated into one number
- ▣ It varies with the strike (smile, skew) and the maturity (term structure): Black–Scholes is a quoting convention, not a model of the data
- ▣ Fat tails, leverage and crash fear all tilt the smile towards low strikes
- ▣ Next: a full surface from a real market, Bitcoin options

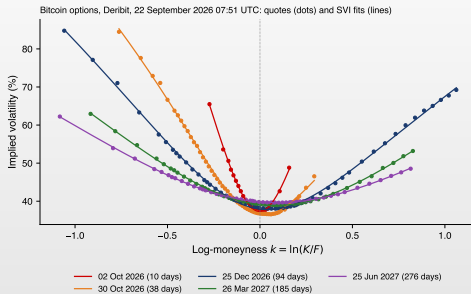
Bitcoin Options on Deribit

- Deribit lists European options on Bitcoin and Ether, traded 24/7
 - ▶ **Inverse** options: premium and pay-off are paid in Bitcoin; the price in USD is the BTC premium times the Bitcoin price
- Daily, weekly, monthly and quarterly expiries; each expiry has its own forward price F
- DVOL: Deribit's 30-day implied-volatility index for Bitcoin [Deribit DVOL](#)
- Our snapshot: 22 September 2026, 07:51 UTC; 9 expiries with at least two days left, 375 quoted OTM options
 - ▶ Cryptos as an asset class and their risk: [Pele et al. \(2023\)](#); hedging Bitcoin options: [Matic, Packham & Härdle \(2023\)](#)

A Parametric Smile: SVI

- Work with the **total implied variance** $w(k) = \sigma_{imp}^2(k) \tau$ at log-moneyness $k = \ln(K/F)$
 - ▶ SVI (stochastic volatility inspired) [Gatheral & Jacquier \(2014\)](#): $w(k) = a + b \left(\rho(k - m) + \sqrt{(k - m)^2 + s^2} \right)$
- Parameters: a level, b slope of the wings, ρ tilt (negative = put skew), m shift, s curvature at the bottom
 - ▶ Wings are linear in k , as required for large strikes
- No arbitrage: call prices must fall and be convex in K (no butterfly arbitrage) and w must increase with τ (no calendar arbitrage)
 - ▶ One SVI fit per expiry by least squares; the convexity condition is tested in Seminar 12, B3

The Bitcoin Smile

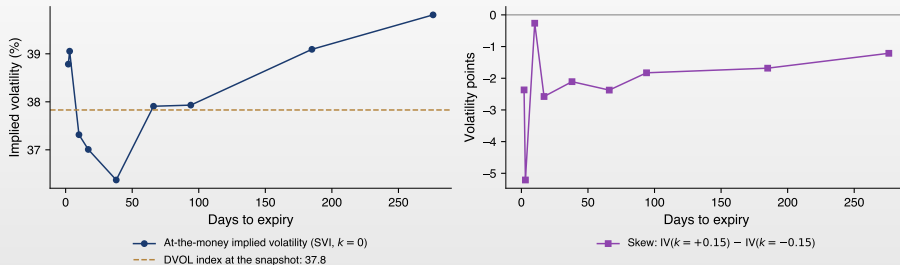


- Mark implied volatilities of OTM options (dots) and SVI fits (lines) for five expiries

Reading the Bitcoin Smile

- ▣ A true **smile**: both wings rise, unlike the one-sided equity skew
 - ▶ Bitcoin can jump up as well as down: call buyers pay for rallies, put buyers for crashes [Zulfiqar & Gulzar \(2021\)](#)
- ▣ Short expiries have the steepest wings: a jump over a few days matters more relative to $\sigma\sqrt{\tau}$
- ▣ SVI fits the quotes closely: root mean squared error at most 0.76 vol points per expiry
- ▣ At this snapshot puts are slightly dearer than calls: skew from -5.2 to -0.3 vol points

Term Structure and Skew



- Left: ATM implied volatility by expiry and the DVOL index; right: skew = implied volatility at $k = +0.15$ minus at $k = -0.15$

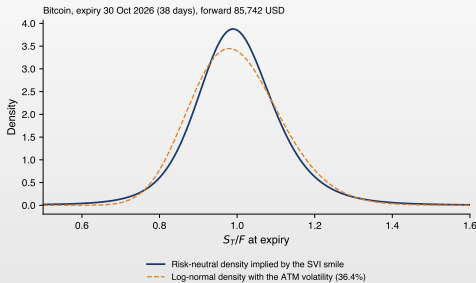
Reading the Term Structure

- ATM implied volatility between 36.4% and 39.8%; the longest expiry (276 days): 39.8%
 - ▶ DVOL 37.8 sits inside the range of the one-month expiries, as it should
- An upward-sloping term structure: the market expects calm now and more uncertainty later
- The skew is negative for 9 of 9 expiries and closest to 0 at long maturities
- A snapshot is one day: the level and the sign of the skew change with the market state

The Risk-Neutral Density

- Breeden & Litzenberger (1978): the second derivative of call prices in the strike gives the risk-neutral density q of S_T
 - ▶ $q(K) = e^{r\tau} \frac{\partial^2 C(K)}{\partial K^2}$; a butterfly spread $C(K-h) - 2C(K) + C(K+h)$ pays like a narrow bet on $S_T \approx K$
- Needs a smooth smile: SVI gives $C(K)$ at every strike, then differentiate twice
- Convexity of C in K (no butterfly arbitrage) is exactly the condition $q \geq 0$
- q mixes probabilities and risk aversion: comparing it with the physical density gives the **pricing kernel** Aït-Sahalia & Lo (1998), Grith, Härdle & Park (2013)

What the Market Prices for Bitcoin



- Risk-neutral density from the SVI smile of the expiry closest to 30 days, vs the log-normal density of Black–Scholes with the ATM volatility

Reading the Risk-Neutral Density

- Expiry in 38 days, forward 85 742 USD, ATM volatility 36.4%
 - ▶ Taller centre and fatter tails than the log-normal density: the smile in density form
- Risk-neutral probability of a fall of more than 20%: 4.3% vs 3.3% under the log-normal density; of a rise above 20%: 5.3% vs 5.3%
- A fall of more than 30%: 1.2% vs 0.1%, about 8 times the log-normal value: the crash tail is where Black–Scholes fails most
- These are prices of insurance, not forecasts: they include the premium for bearing crash risk

Recap: the Volatility Surface

- ▣ A surface = one smile per expiry; SVI gives each smile with five parameters
- ▣ No-arbitrage conditions: convex in the strike, increasing total variance in maturity
- ▣ The smile is the risk-neutral density seen through Black–Scholes
- ▣ Bitcoin: both wings rich, strong term structure, tail probabilities far above the log-normal ones

Variance Swaps

- A **variance swap** pays at T : notional $\times$ (realised variance – strike K_{var}), with realised variance $\frac{252}{n} \sum r_t^2$
 - ▶ The fair strike is the risk-neutral expected variance: no model needed to price it [Demeterfi, Derman, Kamal & Zou \(1999\)](#)
- It can be replicated with a strip of OTM options weighted by $1/K^2$ plus a futures position:
 - ▶
$$K_{var} = \frac{2e^{r\tau}}{\tau} \left(\int_0^F \frac{P(K)}{K^2} dK + \int_F^\infty \frac{C(K)}{K^2} dK \right)$$
- Low strikes get the largest weights: expensive puts make variance expensive
- Buyer of variance = insurance against turbulence; seller = earns a premium, pays in crashes [Carr & Wu \(2009\)](#)

The VIX Index

- Cboe Volatility Index (VIX), since 1993; since 2003 computed from S&P 500 options with the variance-swap formula [Cboe \(2025\)](#)

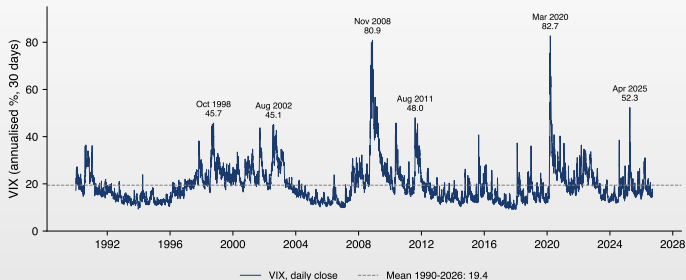
►
$$\sigma^2 = \frac{2}{T} \sum_i \frac{\Delta K_i}{K_i^2} e^{RT} Q(K_i) - \frac{1}{T} \left(\frac{F}{K_0} - 1 \right)^2, \quad Q = \text{mid price of the OTM option at } K_i, \quad K_0 = \text{first strike below } F$$

- Two expiries around 30 days, interpolated to exactly 30 days; $VIX = 100 \sigma$
- VIX futures since 2004, VIX options since 2006 [Cboe \(2025\)](#); the “fear gauge” [Whaley \(2000\)](#)
- Siblings: VIX9D (9 days), VIX3M (3 months), VVIX (the VIX of the VIX)

Worked Example: a VIX for Bitcoin

- Apply the Cboe formula to our Deribit snapshot, with $r = 0$ and USD mid prices
 - ▶ Near expiry: 17 days, 26 OTM options, $\sigma = 39.3\%$
 - ▶ Next expiry: 38 days, 56 OTM options, $\sigma = 40.1\%$
- Interpolate the total variances $\sigma_i^2 T_i$ linearly in time to 30 days:
 - ▶ $\sigma_{30}^2 = \frac{1}{30} \left[T_1 \sigma_1^2 \frac{T_2 - 30}{T_2 - T_1} + T_2 \sigma_2^2 \frac{30 - T_1}{T_2 - T_1} \right]$ (days)
- Our Bitcoin VIX: 39.9; Deribit's DVOL at the same moment: 37.8 [Alexander & Imeraj \(2020\)](#)
- Both exceed the ATM volatility of the two expiries: the variance swap also pays for the wings of the smile

The VIX, 1990–2026

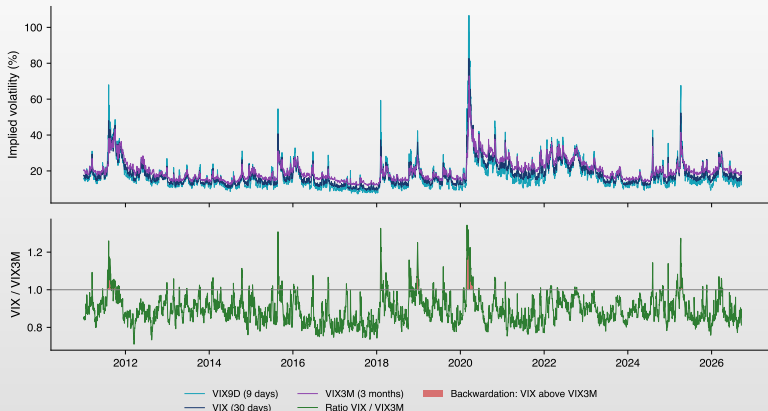


- Daily close with the six highest peaks at least 500 days apart

Reading the VIX History

- ▣ Mean 19.4, median 17.6: an expected annualised volatility of 19.4%, i.e. 1.22% a day or 5.6% a month
 - ▶ Range from 9.1 (2017-11-03) to 82.7 (2020-03-16, COVID-19)
- ▣ Spikes are sudden, declines slow: the same clustering and mean reversion as GARCH volatility (Chapter 5)
- ▣ Above 30 on 7.9% of days, below 15 on 32%: long calm periods, short storms
- ▣ Last value 14.8 (18 September 2026): a calm market by historical standards

The VIX Term Structure

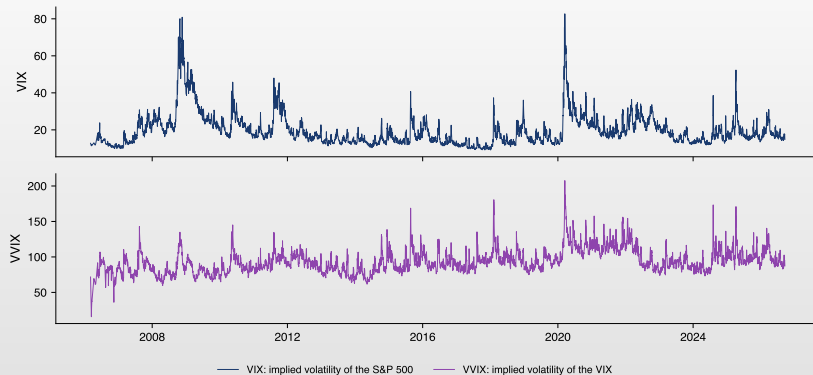


□ Top: implied volatility over 9 days, 30 days and 3 months; bottom: ratio VIX / VIX3M, 2011–2026

Reading the Term Structure

- Normally upward sloping (**contango**): $\text{mean VIX} / \text{VIX3M} = 0.89$; VIX9D below the VIX on 73% of days
 - ▶ In calm markets short-term uncertainty is low and investors pay more for longer protection
- Inverted (**backwardation**) on 7.6% of days, always in crises: fear is concentrated in the near term
- Next 21 days after backwardation: S&P 500 return 2.88% and realised volatility 26.7%, vs 0.80% and 13.6% in contango
- High risk, high expected return after panics: the premium for bearing crash risk [Bollerslev, Tauchen & Zhou \(2009\)](#)

The Volatility of Volatility



□ VIX (top) and VVIX (bottom), daily closes, 2006–2026

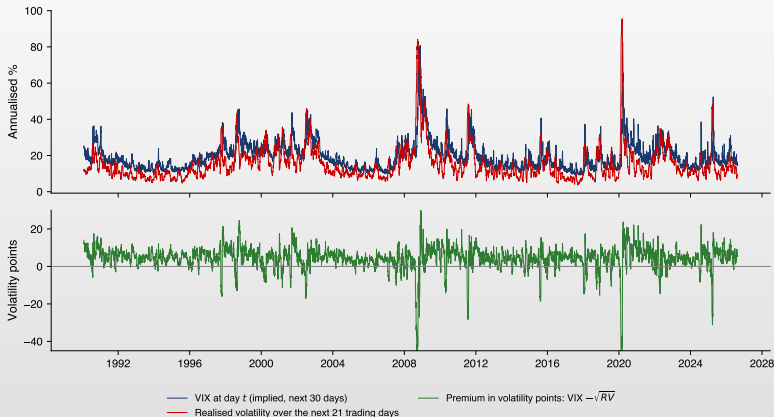
Reading the VVIX

- Mean VVIX 93.5: the VIX itself is expected to move about 5.3 times more than the index, in relative terms
- Highest value 207.6 on 2020-03-16, the same day as the VIX peak
- Daily changes of VIX and VVIX correlate 0.71: when fear rises, uncertainty about fear rises too
- Volatility is itself random: the core of stochastic-volatility models such as Heston [Heston \(1993\)](#)

The Variance Risk Premium

- Variance risk premium (VRP): implied variance minus the variance realised afterwards Carr & Wu (2009), Bollerslev, Tauchen & Zhou (2009)
 - ▶ $VRP_t = VIX_t^2 - RV_{t,t+21}$, with $RV_{t,t+21} = \frac{252}{21} \sum_{i=1}^{21} r_{t+i}^2$ (annualised, in %²)
- It is the average P&L of the seller of a variance swap: the price of insurance against volatility
- Positive on average if investors dislike states with high volatility, which are also bad states for wealth
- Overlapping 21-day windows are autocorrelated: standard errors need Newey–West with 21 lags Newey & West (1987)

Implied vs Realised Volatility, S&P 500

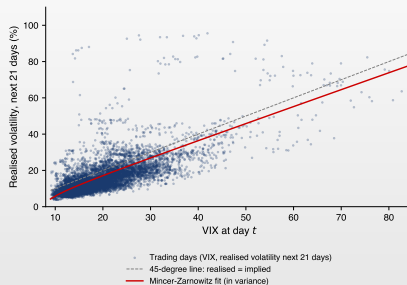


Top: VIX and the volatility realised over the next 21 trading days; bottom: their difference, 1990–2026

Reading the Variance Risk Premium

- ▣ 9 198 days: mean VIX 19.4, mean realised volatility 15.3
 - ▶ Premium 4.09 vol points (Newey–West s.e. 0.27); in variance 113 %², $t = 5.5$
- ▣ Positive on 86% of days: implied volatility is usually too high as a forecast
- ▣ The exceptions are the crashes: worst −72.0 vol points, for the window starting 2020-02-20 (COVID-19)
- ▣ Same message as the delta-hedged option: sellers of volatility are paid on average and lose in crises

Is the VIX a Good Forecast?



- Realised volatility over the next 21 days vs the VIX, with the 45-degree line and the Mincer–Zarnowitz fit [Mincer & Zarnowitz \(1969\)](#)

Reading the Forecast Regression

- $RV_{t,t+21} = \alpha + \beta VIX_t^2 + \varepsilon$: an unbiased forecast has $\alpha = 0$, $\beta = 1$
 - ▶ Estimates: intercept $\hat{\alpha} = -51.7$ (34.4), $\hat{\beta} = 0.86$ (0.10), $R^2 = 0.37$; Newey–West standard errors
- Most points lie below the 45-degree line: the VIX overstates future volatility by the risk premium
- Past realised variance alone explains less: $R^2 = 0.29$; both together 0.38
 - ▶ With both, the VIX keeps its weight (0.77) and past variance adds little (0.08, s.e. 0.13): options already contain the history
- Implied volatility is a biased but informative forecast: useful for risk (Chapters 7–8) once the premium is removed

The Variance Risk Premium of Bitcoin



- DVOL and the Bitcoin volatility realised over the next 30 calendar days, 2021–2026; bottom: their difference

Reading the Bitcoin Premium

- 1975 days: mean DVOL 60.3 vs realised 51.7; premium 8.5 vol points (s.e. 1.6, $t = 5.3$)
 - ▶ Twice the S&P 500 premium in vol points, but similar relative to the level of volatility
- Positive on 71% of days: less often than for the S&P 500; long episodes of negative premium
- Worst: -46.9 vol points for the window starting 2026-01-28
- A young market and a short sample: 66 non-overlapping 30-day periods; a Bitcoin VIX and its premium:
Alexander & Imeraj (2020)

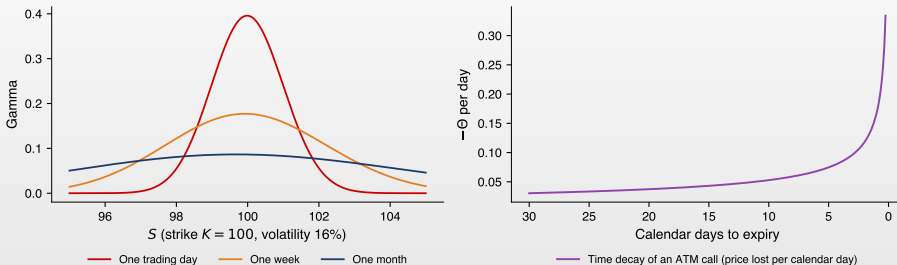
Recap: Implied Volatility as Information

- The VIX is the price of a 30-day variance swap on the S&P 500, built from a strip of options
- Its term structure flags crises (backwardation); VVIX measures uncertainty about volatility
- Implied variance exceeds realised variance on average, for equities and for Bitcoin: the variance risk premium
- As a forecast, the VIX is biased upwards but beats past volatility

Options That Live One Day

- ▣ **0DTE** (zero days to expiry) options: traded on the day they expire [Cboe 0DTE](#)
 - ▶ S&P 500 options now expire every trading day, so a 0DTE contract exists every day
- ▣ More than 75% of retail trades in S&P 500 options involve 0DTE contracts, and retail investors lose money on them [Beckmeyer, Branger & Gayda \(2023\)](#)
 - ▶ 0DTE options were 48% of index-option trading in August 2023 [Brogaard, Han & Won \(2023\)](#)
- ▣ More 0DTE trading is followed by higher index volatility, mostly through speculative retail trades [Brogaard, Han & Won \(2023\)](#); ultra-short pricing needs skewness and kurtosis corrections [Bandi, Fusari & Renò \(2023\)](#)
- ▣ Theory from Section 4: with $\tau \rightarrow 0$ gamma and theta explode at the money

Gamma and Time Decay Near Expiry



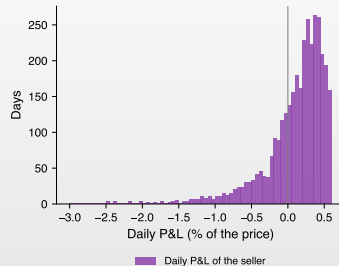
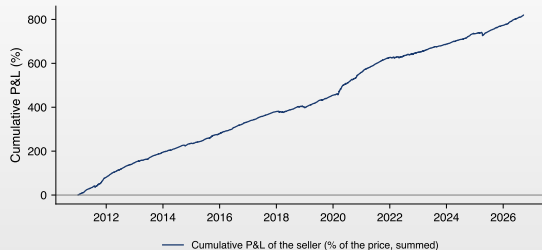
□ Left: gamma of a call with $K = 100$ one day, one week and one month before expiry; right: daily time decay of an ATM call

 MFM_ch12_0dte_gamma

Reading Gamma Near Expiry

- ▣ ATM gamma one day before expiry: 0.396 vs 0.086 one month before: 4.6 times larger ($= \sqrt{21}$)
- ▣ A 1% move in the last day changes delta by about 0.4: the hedge must be adjusted many times within the day
- ▣ Time decay per day: 0.167 in the last day vs 0.030 with 30 days left, about 5.5 times more
- ▣ The seller collects fast time decay and carries large gamma risk: small premium, sudden losses

Selling a Same-Day Straddle: a Stylised Test



- SPY, 2011–2026: sell an ATM straddle at the open, settle at the close; premium = $\sqrt{2/\pi} \sigma_{day} S$ with σ_{day} from the previous VIX9D scaled to the trading session

 MFM_ch12_0dte_straddle

Reading the Straddle Test

- ▣ Session share of daily variance 0.74 (estimated on 1993–2010); 3 950 days
 - ▶ Mean premium 0.76% of the price vs mean absolute open-to-close move 0.55%
- ▣ The seller wins on 75% of days, 20.7 basis points a day on average
- ▣ Worst day -8.25% (2025-04-09), about 11 premia; the move exceeds twice the premium on 3.5% of days
- ▣ Stylised pricing, not traded prices: the result measures the variance premium inside VIX9D
 - ▶ Market 0DTE premia, spreads and intraday hedging would change the numbers [Bandi, Fusari & Renò \(2023\)](#), [Beckmeyer, Branger & Gayda \(2023\)](#)

What We Learned

- ▣ Parity and no-arbitrage bounds need no model; the tree and Black–Scholes price by replication
- ▣ The Greeks measure exposures; discrete hedging leaves an error of order $1/\sqrt{N}$
- ▣ A delta-hedged option is a bet on realised vs implied variance; sellers earned a premium on the S&P 500 and on Bitcoin
- ▣ Implied volatility varies with strike and maturity: fat tails, leverage and crash fear shape the surface
- ▣ The VIX turns the whole smile into one number, a biased but informative forecast of volatility

Which Tool When?

Question	Tool
Is this quote consistent with the others?	Put–call parity, no-arbitrage bounds
Price an American option	Binomial tree (CRR) with early exercise
Hedge a book of options	Greeks; delta-gamma hedging (Seminar 12, A6)
Interpolate quotes, check arbitrage	SVI per expiry; convexity and calendar checks
What does the market fear?	Risk-neutral density, skew, VIX term structure
Forecast volatility for risk	VIX or DVOL minus the variance risk premium, with GARCH (Chapter 5)

Coming Next and Further Reading

- Chapter 13: machine learning for financial prediction, with volatility and option-implied features among the inputs
- Options and the implied-volatility surface: Franke, Härdle & Hafner (2019), Härdle, Chen & Overbeck (2017), Čížek, Härdle & Weron (2011), Gatheral (2006)
- Pricing kernels from option prices: Aït-Sahalia & Lo (1998), Grith, Härdle & Park (2013)
- Crypto options and their risk: Matic, Packham & Härdle (2023), Alexander & Imeraj (2020), Pele et al. (2023)

Further Study on Quantinar

Advanced courses on quantinar.com

-  XFG Advanced Methods in Quantitative Finance
-  Pricing Kernels and Risk Premia
-  Hedging Cryptocurrency Options

Code and data

- Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_12

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